VOLATILITY_INSTRUMENTS

NOTE: CALCULUS IS THIS ONE . We need to fgind the discrete ficnacnial caluclus pdf byut Frogy eithr online or locally

PAYOFF

Definition 1 (Volatility option). Fix a strike variance σ_K^2. The **volatility option** with vega notional ΔQ_v is the contract paying

$$\pi^{\sigma} \; = \; \Delta Q_v \left(\sigma^2 \left(i(t) \right) - \sigma_K^2 \right)^{+}$$

where $\sigma^2(i(t))$ is the realized tick variance. Consequently $\Delta Q_v \equiv \Delta\pi^{\sigma} / \Delta\big(\sigma^2(i(t)) - \sigma_K^2\big)^{+}$: the notional **is** the option's vega.

Convention 2 (Volatility tick argument). Volatility always takes a **tick argument**: $\sigma^2(i(t))$ is the variance along the tick path at calendar time t, $\sigma^2(i(T))$ its value at the horizon T, and $\sigma^2(i_K) \equiv \sigma_K^2$ the strike variance at the strike tick — the subscript form is declared shorthand for the tick-argument form, as is $\sigma_R^2(T) \equiv \sigma^2(i(T))$. A bare σ^2 is not well-formed. *(Adopted from the converted region upward; the sections below are swept as the pair pass reaches them.)*

Settlement form of Definition 1. At unit notional, the contract settles on realized variance at the horizon:

$$\pi^{\sigma} \left(\sigma_K, T; t \right) \; = \; \left(\sigma^2(i(T)) - \sigma^2(i_K) \right)^{+}$$

Following VOL_SWAPS, the price of the volatility option is the *cost of replicating it with options*. This is where panoptic enters. The replication proved in-tree is the **ladder** form (the $\xi^{\star}$ log-contract weights, `variancePortfolio_upsilon`); whether it collapses to a **two-instrument** affine form $p_{\pi^{\sigma}} \; = \; p_0 + a_1 \, p_{\pi^{\mathrm{call}}} + a_2 \, p_{\pi^{\mathrm{put}}}$ is **OPEN** — statement parked pending the liquidity-side definitions, per the 12.1 ledger.

Definition 2 (Theta). The **theta** of the call (resp. put) at strike tick i_K is the per-time-step payoff variation

$$\theta \left(\, p_{(\eta,\Delta_i)} \left(i; t \right), p_{(\eta,\Delta_i)} \left(i_K \right), \sigma \left(i(t) \right) \right) \; \equiv \; \frac{\Delta\pi^{\mathrm{call} \,|\, \mathrm{put}}}{\Delta \, t}$$

The strike is the price at the STRIKE TICK i_K, on the price grid $p_{(\eta,\Delta_i)}$ defined under the pricing geometry below (`VolInstrument.priceEta`).

Proposition 2 (Closed form of θ). Under the price grid $p_{(\eta,\Delta_i)}$,

$$\theta \; = \; \frac{p_{(\eta,\Delta_i)} \left(\cdot \right) \sigma \left(i(t) \right)}{\sqrt{8 \, \pi \, t}} \; \exp \left(- \frac{\left[- \ln \left(\frac{p_{(\eta,\Delta_i)} \left(i(t_0) \right)}{p_{(\eta,\Delta_i)} \left(i_K \right)} \right) + \frac{\sigma^2(i(t)) \, t}{2} \right]^2}{2 \, \sigma^2(i(t)) \, t} \right)$$

Rule 1 (Option pricing). The protocol prices the call (resp. put) at strike tick i_K as **accumulated theta along the realized tick path**:

$$p_{\pi^{\mathrm{call} \,|\, \mathrm{put}}} \left(t \right) \; \leftarrow \; \int_{t_0}^{t} \theta \left(\, p_{(\eta,\Delta_i)} \left(i; s \right), p_{(\eta,\Delta_i)} \left(i_K \right), \sigma \left(i(s) \right) \right) \varsupsetneqq s$$

The left arrow marks a Rule, not an identity: this is a stipulation of the protocol, and the implementation either complies with it or does not.

RESOLVED (user ruling, 2026-08-03): exponent sign is **NEGATIVE**. Two-part justification: (i) with the display's own prefactor $p(\cdot)\sigma/\sqrt{8\pi t}$, the bracket is $-\sigma\sqrt{t}\,d_2$, so the negative sign gives $e^{-d_2^2/2} \propto \varphi(d_2)$ — exactly the $r = 0$ Black–Scholes dt-leg $\theta = S\sigma\varphi(d_1)/(2\sqrt{t})$ via $S\varphi(d_1) = K\varphi(d_2)$; (ii) DECISIVE and internal: $p_{\pi^{\text{call|put}}} \leftarrow \int \theta$ over the price grid CONVERGES only with the negative sign (Gaussian tails) — under $+$ the assignment defining the option prices diverges. The ATM form cannot discriminate (`theta_atm_closed_form`, exponent vanishes ATM); the tails do.

Definition 4 (Upsilon). The **upsilon** of a premium or payoff functional at strike tick i_K is its per-unit-variance sensitivity, as a lattice finite difference in the variance argument:

$$\upsilon\left(p_{(\eta,\Delta_i)}\,(i;t)\,,\,p_{(\eta,\Delta_i)}\,(i_K)\right) \;\equiv\; \frac{\Delta\pi^{\text{call | put}}}{\Delta\sigma^2\,(\cdot)}$$

Proposition 3 (Vega bridge). On the region where the volatility option is in-the-money at both variance endpoints ($\sigma^2(i_K) \le \sigma^2(i(t))$ and $\sigma^2(i_K) \le \sigma^2(i(t)) + \Delta s$), its upsilon **is** its vega notional:

$$\upsilon(\pi^\sigma) \;=\; \Delta Q_\upsilon$$

— the dimensional bridge the identification sought: υ occupies the ΔQ_υ slot ($= \;\Delta Q_M/p_{\text{risk}}$ via `Flow.deltaShares`). Off that region the recovery FAILS by construction (the kink); the ATM/OTM null is recorded as a conjecture, unproven.

Definition 5 (Replicating portfolio). The **replicating portfolio** $\Pi^{\text{call|put}}(\sigma; p_{(\eta,\Delta_i)}(i;t))$ is the option portfolio whose sensitivity to realized variance is independent of the underlying price PG7 — a single option cannot serve, since a price move alters its variance sensitivity.

Convention 1 (Replication relation). For payoff claims A, B we write $A \equiv^R B$ — "A **is replicated by** B" — when B's payoff reproduces A's. This is a *claim about two objects*, not a definitional identity: each instance must be proved, and until it is, it is stated OPEN. (This is the relation the two-instrument question above is posed in.)

Definition 6 (Log portfolio). For $p^\star$ the approximate at-the-money forward level marking the boundary between liquid puts and liquid calls PG9, the **log portfolio** and its running form are

$$\Pi^{\text{call|put}}\big(\sigma; p_{(\eta,\Delta_i)}(i;t)\big) \;=\; \frac{p_{(\eta,\Delta_i)}(i;t) - p^\star}{p^\star} \;-\; \log\Big(\frac{p_{(\eta,\Delta_i)}(i;t)}{p^\star}\Big) \;+\; \frac{\sigma^2(i(t))\,t}{2}$$

(at $t = 0$ the running term vanishes; $\Pi \ge 0$ with $\Pi(p^\star) = 0$). Its **unit-vega normalized form** (Theorem 3's Id_{N_σ}), with the remaining-variance tail:

$$\Pi^{\text{call | put}}\,(\sigma; p_{(\eta,\Delta_i)}\,(i;t); T) \;=\; \mathrm{Id}_{N_\sigma}\Big[\frac{p_{(\eta,\Delta_i)} - p^\star}{p^\star} \;-\; \log(\frac{p_{(\eta,\Delta_i)}}{p^\star})\Big] \;+\; \frac{T-t}{T}\,\sigma^2(i(t))$$

Settlement instantiation (Convention 2): $\Pi^{\text{call|put}}\big(\sigma^2(i(T));\; p_{(\eta,\Delta_i)}(i;t);\; T\big) \;=\; \mathrm{Id}_{N_\sigma}\Big[\frac{p_{(\eta,\Delta_i)}-p^\star}{p^\star} \;-\; \log(\frac{p_{(\eta,\Delta_i)}}{p^\star})\Big] \;+\; \frac{T-t}{T}\,\sigma^2(i(t))$.

Proposition 4 (Ladder replication). The volatility option is replicated by the log portfolio:

$$\pi^\sigma(t) \;\equiv^R\; \Pi^{\text{call|put}}\big(\sigma; p_{(\eta,\Delta_i)}(i;t)\big)$$

and Π has Definition 5's defining property: its variance sensitivity is **constant in the underlying price**, $\upsilon(\Pi) = T/2$.

Status: the $\equiv^R$ core is **adapted from the variance-swap text** Demeterfi and is **OPEN in-tree** — Convention 1's discipline applies. PROVED: the sensitivity half. OWED: the payoff-reproduction step connecting `variancePortfolio` to `volOptionPayoff` — an Aristotle target.

Rule 3 (Ladder allocation). The protocol realizes the log portfolio on the grid as the strike ladder — the weight profile $\ell(\xi^\star, \iota; i_K)$ being a geometric liquidity distribution in the sense of the Bunni v2 whitepaper (whose general LDFs $\ell_{\mathrm{LDF}}(\theta_{\mathrm{LDF}}; i_K)$ are the declared FUTURE MILESTONE, G4):

$$\Pi^{\mathrm{call|put}}\big(\sigma; p_{(\eta,\Delta_i)}(i;t)\big) \;\leftarrow\; \sum_{i_K} L(i_K)\,\Pi^{\mathrm{call|put}}\big(\sigma_K; p_{(\eta,\Delta_i)}(i;t)\big), \qquad L(i_K) = \bar{L}\,\ell(\xi^\star, \iota; i_K)$$

The left arrow marks the Rule: an **allocation the protocol enforces**, not an equality — $\Pi^{\mathrm{call|put}}(\sigma_K; \cdot)$ is the per-strike member at strike tick i_K, and L is Definition 7's ladder. Whether the enforced ladder's payoff reproduces the log contract is part of Proposition 4's OPEN core, not asserted here.

Definition 7 (Liquidity ladder). Per strike tick i_K, the ladder allocates the total liquidity $\bar{L}$ by the geometric weight profile

$$L\,(i_K) \;=\; \bar{L}\,\ell\,(\xi, \iota; i_K), \qquad \bar{L} \;=\; \sum_{i_K = i_{\min}}^{i_{\max}} L\,(i_K), \qquad \ell\,(\xi, \iota; i_K) \;=\; \frac{\xi^{i_K}}{\left(\frac{1-\xi^{\iota}}{1-\xi}\right)}$$

with ladder parameter set $\Theta_\ell = \{\xi, \iota\}$ — see **PROTOCOL_PARAMETERS (Θ_ℓ)**.

Theorem 2 (Partition of unity). The weights are a partition of unity and the δ-neutral ratio is pinned:

$$\sum_{i_K} \ell\,(\xi, \iota; i_K) \;=\; 1, \quad \ell > 0\;(\xi \in (0,1) \cup (1, \infty)), \quad \lim_{\xi \to 1} \ell \;=\; \frac{1}{\iota}$$

PROTOCOL_PARAMETERS

Every parameter of the protocol enters here as a **Protocol Parameter** — a special definition that fully specifies its **domain**, its **purpose**, and its **economic meaning**, indexed by its parameter set. A parameter not listed here is not a parameter of the protocol.

Protocol Parameter ($\Theta_\ell = \{\xi, \iota\}$ — the ladder).

- ξ — the **liquidity ratio**. *Domain:* $\xi \in (0,1) \cup (1, \infty)$; $\xi = 1$ is reached by limit only (Theorem 2). *Purpose:* sets the geometric decay of per-strike liquidity in the ladder (Definition 7); with ι, encodes the strike weights that make the portfolio **delta-neutral**. *Economic meaning:* the ratio of liquidity between adjacent strikes; pinned at $\xi^\star = \lambda^{-\Delta_i/2}$, the log-contract weight law under which the ladder replicates the variance payoff (Proposition 4).
- ι — the **ladder resolution**. *Domain:* $\iota \in \mathbb{N}$, $\iota \geq 1$. *Purpose:* the number of strikes carrying the ladder (Definition 7); the weight profile lives on the simplex $\Delta^{\iota-1}$; with ξ, encodes the **delta-neutral** strike weighting. *Economic meaning:* the resolution at which the continuous log-contract strip is discretized — the finite-strip replication error and the G4 underspecification deficit ($\iota - 2$) are both functions of it.

Protocol Parameter ($\Theta_p = \{\eta, \Delta_i\}$ — the pricing geometry).

- η — the **grid exponent**. *Domain:* $\eta > 0$ (jointly with $\Delta_i > 0$ this is exactly the strict-monotonicity hypothesis $\eta\,\Delta_i > 0$, `priceEta_strictMono`; $\eta = 1$ is the canonical grid). *Purpose:* the one-parameter deformation of the tick-price law (Definition 8) — the exponent tilting the grid away from the square-root-price member. *Economic meaning:* the grid-side tilt dial. It is **not** the trading-curve share: η enters the curve only through the proven bridge $\chi_{X/M}(\eta) = \Lambda(\eta\, \Delta_i \ln \lambda/2)$, and the genuine curvature $\varkappa_\varphi$ does not depend on it at all (a function of $\epsilon_{X/M}$ alone).

- Δ_i — the **tick spacing**. *Domain:* $\Delta_i > 0$ (the Lean leg-nonnegativity theorem needs $\Delta_i \geq 0$ in addition to $\eta\, \Delta_i > 0$; on-chain it is the positive integer tick spacing of UNI_V3). *Purpose:* grid granularity — the quantization step at which strikes, hence ladder legs, may sit (Definition 8). *Economic meaning:* the spacing pins the ladder ratio $\xi^\star = \lambda^{-\Delta_i/2}$ (Θ_ℓ entry) and sets the per-spacing price step $\lambda^{\eta\Delta_i/2}$ — the coarseness lever coupling the pricing geometry to the replication ladder.

  On the grid, η and Δ_i are REDUNDANT — they enter only through the product $\eta\Delta_i$ (Theorem 21); they separate off-grid ($\xi^\star$, Proposition 6).

Protocol Parameter ($\Theta_\varphi = \{\chi_{X/M}, \epsilon_{X/M}\}$ — the trading curve).

- $\chi_{X/M}$ — the **share parameter**. *Domain:* $\chi_{X/M} \in (0, 1)$ (Definition 12). *Purpose:* the exponent weighting the ΔQ_M leg of the trading function (Definition 12); first slot of the subscript tuple $(\chi_{X/M}, \epsilon_{X/M})$. *Economic meaning:* the SHARE (distribution) parameter — the fraction of pool value held in the ΔQ_M leg; it says WHERE the value sits. $\chi_{X/M} = 1/2$ is the balanced pool; moving it tilts inventory toward one leg WITHOUT changing how the curve resists trade. Via the proven bridge $\chi_{X/M}/(1 - \chi_{X/M}) = \lambda^{\eta\Delta_i/2}$ it is an **observable of the price grid**, not an independent primitive — subject to the OPEN leg-orientation FLAG (Definition 12), which flips the bridge.
- $\epsilon_{X/M}$ — the **substitution parameter**. *Domain:* $\epsilon_{X/M} \in (-\infty, 1]$ — $\epsilon_{X/M} = 1$ the linear member (perfect substitutes, $\bar{\epsilon}_{X/M} = \infty$); $\epsilon_{X/M} = 0$ the defined Cobb–Douglas case (constant product); $\epsilon_{X/M} \to -\infty$ the Leontief limit (no trade). *Purpose:* the substitution axis of Definition 13, second slot of the subscript tuple; the elasticity of substitution is $\bar{\epsilon}_{X/M} = 1/(1 - \epsilon_{X/M})$, and the genuine curvature $\varkappa_\varphi$ is a function of this axis ALONE. Proven orthogonal to the share axis (`phiCES_rho_ne_eps_axis`; $\varsigma_{X/M}$ factors through the share, `curvIndex_is_rho_zero_slice`). *Economic meaning:* the slippage dial — HOW HARD the pool resists being moved. This is what an arbitrageur pays for: less substitutability means more price impact per unit extracted — and equally worse execution for the ordinary investor, which is why both effects move together and produce an interior optimum.

Protocol Parameter ($\Theta_\phi = \{\gamma, \bar{\phi}, \beta, \alpha\}$ — the fee schedule).

- $\bar{\phi}$ — the **fee floor**. *Domain:* $\bar{\phi} \geq 0$. *Purpose:* the unconditional base of the schedule (Definition 18). *Economic meaning:* LPs take a base fee at every volatility — the schedule never degenerates to free execution (Theorem 1's lower envelope).
- $\alpha = \{\alpha_j, \alpha_R\}$ — the **surcharge scales**. *Domain:* $\alpha_j \geq 0$, $\alpha_R \geq 0$. *Purpose:* scale each sigmoid's maximum (ALGEBRA eq. (4)); $\sum_j \alpha_j$ times the gate ceiling α_R sets the width of Theorem 1's fee band. *Economic meaning:* the volatility surcharge budget — what heavy trading in volatile conditions can add above the floor.
- $\beta = \{\beta_j, \beta_R\}$ — the **transition midpoints**. *Domain:* real. *Purpose:* place each sigmoid's transition; they position the ramp *inside* the band without moving its edges. *Economic meaning:* the volatility (resp. utilization) levels at which the surcharge switches on — G3's placement-not-level reading.
- $\gamma = \{\gamma_j, \gamma_R\}$ — the **steepnesses**. *Domain:* $\gamma_j > 0$ (Theorem 1's monotonicity hypothesis). *Purpose:* the ramp steepness (single-term case: $s_f = 1/\gamma_0$). *Economic meaning:* how sharply the schedule reacts near its midpoint — the dial between smooth repricing and a near-step surcharge.

  Parameter registry COMPLETE: Θ_ℓ, Θ_p, Θ_φ, Θ_ϕ. The former Θ_{ord} is NOT a parameter set — it is user-supplied per order and lives as $\mathcal{I}_{\text{ord}}$ under **# PROTOCOL_INPUTS** (the third registry class).

PROTOCOL_CONSTANTS

Every fixed numeral of the protocol enters here as a **Protocol Constant** — a value the protocol fixes once, **not a design dial**: it belongs to no $\Theta_{\bullet}$, and no statement may treat it as free. Indexed by its constant set.

Protocol Constant ($\mathcal{C}_p = \{\lambda\}$ — the pricing geometry).

- λ — the **tick base**. *Value:* $\lambda = 1.0001$ (UNI_V3). *Purpose:* the base of the price grid (Definition 8); every grid ratio in the document — $\lambda^{-\Delta_i}$ (Theorem 4), $\xi^\star = \lambda^{-\Delta_i/2}$ (Proposition 6, Θ_ℓ) — is a power of it. *Economic meaning:* one tick = one basis point of price — the minimal price quantum of the underlying market.

Proposition 5 (Single-leg direction sensitivity). A single leg's variance sensitivity is direction-sensitive:

$$\frac{\Delta\pi^{\text{call | put}}}{\Delta\,\sigma} \approx \frac{\Delta\theta}{\Delta\sigma}$$

inheriting the sign of $\ln\left(p_{(\eta,\Delta_i)}(i;t)/p_{(\eta,\Delta_i)}(i_K)\right)$ — a single option cannot carry Definition 5's price-independence, which is why the ladder exists.

Status: **OPEN** — pinned in-tree as `Upsilon.ATMOTMNullHypothesis`, a Prop **conjecture, no proof, no axiom**. One correction is machine-recorded and travels with it: the naive strike-centered envelope $e^{-c|i-i_K|}$ is **FALSE on the entire left branch for every** $c > 0$ (the forward difference is right-shifted; a parameter-independent obstruction) — the honest envelope is centered on the peak pair $\{i_K{-}1, i_K\}$. The conjecture's originally named test avenue (the econometric track) is CLOSED-terminal, so it either gets a formal proof or stays open.

Theorem 3 (Unit vega).

$$\frac{\Delta\,\Pi^{\text{call | put}}(\cdot)}{\Delta\,\sigma^2}N_\sigma \;=\; T/2\,N_\sigma \;\Rightarrow\; \mathrm{Id}_{N_\sigma} \;\equiv\; \frac{2}{T}$$

Theorem 13 (Maturity equivalence). *(Moved here from # PROTOCOL_INPUTS, user ruling 2026-08-04 — both premises are this section's results; $\Delta Q_\upsilon^\star$ is the target-vega Protocol Input, # PROTOCOL_INPUTS.)* From $\upsilon = T/2$ (`variancePortfolio_upsilon`) and $\mathrm{Id}_{N_\sigma} = 2/T$ (`variancePortfolio_unit_upsilon`):

$$T^\star \;=\; 2\,\frac{\Delta Q_\upsilon^\star}{N_\sigma} \quad\Longleftrightarrow\quad \Delta Q_\upsilon^\star \;=\; \frac{T^\star}{2}\,N_\sigma$$

with $\Delta Q_\upsilon^\star, N_\sigma > 0 \;\Rightarrow\; T^\star > 0$, and $T^\star$ strictly increasing in $\Delta Q_\upsilon^\star$, strictly decreasing in N_σ. The perpetual order specifies no T; $T^\star$ is the implied maturity of the equivalent dated variance contract — derived from $\Delta Q_\upsilon^\star$, never stored.

Rule 4 (Position ledger). The protocol books a position as its **net signed liquidity** per strike: each leg carries the direction sign $\mathbb{I}_{\text{long|short}}$, and the ladder's ledger is

$$\pi^\sigma(\sigma_K, T; t) \leftarrow \sum_{i_K} L(i_K)\,\mathbb{I}_{\text{long|short}}, \qquad \mathbb{I}_{\text{long|short}} \equiv \begin{cases} -1 & \text{long (liquidity removed — burn)} \\ 1 & \text{short (liquidity minted)} \end{cases}$$

The sign is **per leg** (`isLong` in the Panoptic tokenId), so mixed-direction ladders are expressible. Leg **type** is not an index here: put or call is determined structurally by the strike against $p^\star$ — puts below, calls above (Definition 6) — and is carried by `tokenType`.

PRICING_GEOMETRY

Definition 8 (Price grid). The **price grid** is the map assigning to each tick i the value

$$p_{(\eta,\Delta_i)}(i) \;\equiv\; \lambda^{i/2\,\Delta_i\,\eta}$$

where λ is the fixed tick base — a **Protocol Constant** (see **PROTOCOL_CONSTANTS ($\mathcal{C}_p$)**), not a member of Θ_p — and $(\eta, \Delta_i) = \Theta_p$ are Protocol Parameters (see **PROTOCOL_PARAMETERS (Θ_p)**). At $\eta = 1$ the grid is the canonical square-root-price tick law of UNI_V3 (`priceEta_one`); the strike price of Definition 1 is the grid at the strike tick, $p_{(\eta,\Delta_i)}(i_K)$.

η (price grid) and $\chi_{X/M}$ (trading curve, $\varphi_{(\chi_{X/M},0)}$) are DISTINCT parameters on distinct objects; they are not two names for one exponent. Their relation is a THEOREM, not a definition — see the $\chi_{X/M} \leftrightarrow \eta \leftrightarrow \varsigma_{X/M}$ block.

Theorem 21 (Half-kernel factorization: rescaling and partition change). Definition 8's grid factors through the canonical geometry in two ways.

(i) **Rescaling:** $p_{(\eta,\Delta_i)} = p_{(1,\eta\Delta_i)}$ — on the grid, η and Δ_i enter ONLY through the product $\eta\Delta_i$; the grid alone cannot identify them separately (they separate off-grid: $\xi^\star = \lambda^{-\Delta_i/2}$, Proposition 6, depends on Δ_i alone).

(ii) **Partition change (the pricing-implementation theorem):** for ANY admissible (η, Δ_i) and any reference spacing $\bar{\Delta}_i \neq 0$, the price is a PRODUCT of two canonical-geometry prices whose tick arguments are functions of the current tick:

$$p_{(\eta,\Delta_i)}(i) \;=\; p_{(1,\bar{\Delta}_i)}(i^\star) \cdot p_{(1,\bar{\Delta}_i)}(i^\circ), \qquad i^\star = i^\circ = \frac{i\,\Delta_i\,\eta}{2\,\bar{\Delta}_i}$$

exactly on integer ticks under the commensurability $(i^\star + i^\circ)\,\bar{\Delta}_i = i\,\Delta_i\,\eta$; an Int24-windowed split with witnesses $i_- = \lfloor \eta\, i \rfloor$, $i_+ = i - i_-$ realizes it inside the Uniswap/Plank tick domain. This is why the $\frac{1}{2}$ sqrt-price algebra CLOSES under η — the plank implementation prices every η member using only canonical-kernel evaluations. *Convention bridge:* the exp layer states these on its $\bar{\eta}$-kernel, whose canonical member is written $\bar{\eta} = 1/2$; by T28'a's factor two that member IS Definition 8's $\eta = 1$ grid, and the identity makes no factor-share identification.

Theorem 4 (Geometric strike-notional weights). On the price grid $\lambda^{i\,\Delta_i}$ — the square of Definition 8's grid at $\eta = 1$ (`priceGrid_eq_tickPrice_sq`) — the discretized strike-notional weights of the log contract are exactly geometric:

$$\frac{\lambda^{(i+1)\Delta_i} - \lambda^{i\,\Delta_i}}{\left(\lambda^{i\,\Delta_i}\right)^2} \;=\; (\lambda^{\Delta_i} - 1)\left(\lambda^{-\Delta_i}\right)^i$$

with ratio $\lambda^{-\Delta_i}$ — λ the fixed tick base (Protocol Constant $\mathcal{C}_p$), so the ratio is a function of the protocol parameter Δ_i alone. Normalized, the weights are the geometric profile at that ratio, and Theorem 2's partition of unity applies.

Proposition 6 (The liquidity ratio $\xi^\star$). The per-tick *liquidity* replicating the log contract scales as the inverse square root of that grid, hence

$$\xi^\star \;=\; \lambda^{-\Delta_i/2} \quad \text{(NOT } \lambda^{-\Delta_i}\text{; the two differ by the tranche-gamma Jacobian)}$$

λ being fixed ($\mathcal{C}_p$), $\xi^\star$ is pinned by Δ_i alone — consistent with the Θ_ℓ registry entry, where ξ is the parameter and $\xi^\star = \lambda^{-\Delta_i/2}$ its pinned value.

Status: the sampling half is **proved** — $K^{-1/2}$ on the grid is geometric with ratio $\lambda^{-\Delta_i/2}$ (`logContractLiquidity_geometric`). The replication premise — $\ell(K) \propto K^{-1/2}$ from the curvature relation $\ell(P) = -2P^{3/2}V''(P)$, $V''(P) = -1/P^2$, adapted from VOL_SWAPS — is **OPEN** in-tree (the payoff-level curvature bridge is future work).

TRADING_REGION

Definition 9 (Per-strike amounts). On the underlying market (X, M), the liquidity $L(i_K)$ at strike tick i_K holds the per-strike token amounts

$$\Delta Q^L_M(i_K) \equiv L(i_K)\Big[\frac{p_{(\eta,\Delta_i)}(i_K+\Delta_i) - p_{(\eta,\Delta_i)}(i_K)}{p_{(\eta,\Delta_i)}(i_K)\, p_{(\eta,\Delta_i)}(i_K+\Delta_i)}\Big]$$

$$\Delta Q^L_X(i_K) \equiv L(i_K)\Big[p_{(\eta,\Delta_i)}(i_K+\Delta_i) - p_{(\eta,\Delta_i)}(i_K)\Big]$$

identical to the per-rick amounts of BUNNI_V2 §2.3, eqs. (10)–(13), at $\eta = 1$ — stated here on the general grid $p_{(\eta,\Delta_i)}$ (Definition 8). $M \leftrightarrow \text{token}_0$, $X \leftrightarrow \text{token}_1$. **PR-REGION OPEN:** the legs are stated unsigned; the admissibility region of signed flows is not yet defined — Theorem 5's $\Delta_i \geq 0$ hypothesis currently stands in for it.

Theorem 5 (Leg nonnegativity, reciprocal money leg). Nonnegativity of both legs requires $\Delta_i \geq 0$ in addition to $\eta\,\Delta_i > 0$ ($\eta, \Delta_i < 0$ makes $i_K + \Delta_i < i_K$ and reverses signs), and the money leg is the reciprocal-price difference:

$$0 \leq L,\ \eta\,\Delta_i > 0,\ \Delta_i \geq 0 \ \Rightarrow\ \Delta Q^L_M, \Delta Q^L_X \geq 0; \qquad \Delta Q^L_M(i_K) = L(i_K)\Big[\frac{1}{p_{(\eta,\Delta_i)}(i_K)} - \frac{1}{p_{(\eta,\Delta_i)}(i_K+\Delta_i)}\Big]$$

The reciprocal form is exactly BUNNI_V2 eq. (10)'s shape.

Definition 10 (Cumulative amounts). The **cumulative amounts** aggregate the per-strike amounts from the money side down and the asset side up:

$$Q^L_M(i_K) \equiv \sum_{i=i_K}^{i_{\max}} \Delta Q^L_M(i)$$

$$Q^L_X(i_K) \equiv \sum_{i=i_{\min}}^{i_K} \Delta Q^L_X(i)$$

identical to the cumulative amount functions of BUNNI_V2 §2.3, eqs. (14)–(15).

Definition 11 (Inverse cumulative amounts). The **inverse cumulative amounts** map a target amount back to the extremal strike tick attaining it:

$$Q^L_M(\bar{Q}_M)^{-1} \equiv \arg\max_i \Big\{Q^L_M(i_K) : Q^L_M(i_K) \geq \bar{Q}_M\Big\}$$

$$Q^L_X(\bar{Q}_X)^{-1} \equiv \arg\min_i \Big\{Q^L_X(i_K) : Q^L_X(i_K) \geq \bar{Q}_X\Big\}$$

identical to the inverse cumulative amount functions of BUNNI_V2 §2.4, eqs. (22)–(23) — including the arg max/arg min asymmetry, which mirrors the opposed summation directions of Definition 10.

Theorem 6 (Monotonicity and telescoping). Both cumulatives are monotone in the step count (for $L \geq 0$), so the inverse cumulatives of Definition 11 are well-defined least attaining steps; for the constant ladder $L \equiv \bar{L}$ they telescope to closed form:

$$Q_X^L = \bar{L}\left[p_{(\eta,\Delta_i)}(i_{\min} + n\Delta_i) - p_{(\eta,\Delta_i)}(i_{\min})\right], \qquad Q_M^L = \bar{L}\left[\frac{1}{p_{(\eta,\Delta_i)}(i_{\min})} - \frac{1}{p_{(\eta,\Delta_i)}(i_{\min} + n\Delta_i)}\right]$$

Notation map (Bunni v2 remap — collision-driven). BUNNI_V2's symbols do not enter this document because a_0, a_1 collide with the replication weights a_1, a_2 (Settlement form of Definition 1). The remap: $a_0 \longmapsto \Delta Q_M^L$, $a_1 \longmapsto \Delta Q_X^L$, $A_0 \longmapsto Q_M^L$, $A_1 \longmapsto Q_X^L$, $A_0^{-1} \longmapsto Q_M^L(\cdot)^{-1}$, $A_1^{-1} \longmapsto Q_X^L(\cdot)^{-1}$, $w \longmapsto \Delta_i$, $r \longmapsto i_K$, $l_r \longmapsto L(i_K)$, $1.0001 \longmapsto \lambda$ ($\mathcal{C}_p$). Structural identification: Definition 7's weight profile ℓ **is** Bunni's geometric LDF $d_{\alpha,l}$ (§2.2.1, eq. (9)) under $\alpha \longmapsto \xi$, $l \longmapsto \iota$, and Definition 7's $L(i_K) = \bar{L}\,\ell(\cdot)$ is his $l_r = L \cdot LDF_w(r)$ (eq. (5)).

Definition 12 (Weighted-geometric-mean trading function). For share parameter $\chi_{X/M} \in (0,1)$, the **trading function** at strike tick i_K takes as exogenous a trading flow $\Delta Q = (\Delta Q_M, \Delta Q_X)$ and returns

$$\varphi_{(\chi_{X/M},0)}(i_K; \Delta Q, L) \equiv \left(\Delta Q_M^L(i_K) + \Delta Q_M\right)^{\chi_{X/M}} \cdot \left(\Delta Q_X^L(i_K) + \Delta Q_X\right)^{1-\chi_{X/M}}$$

The flow is **exogenous** — trade legs arriving against the endowed per-strike amounts of Definition 9: the endowments are the state, the flow is the input. It is a **trading function** in the sense of CFMM_ GEOMETRY, already in canonical form (nondecreasing, concave, homogeneous); its logarithm is the weighted logarithmic utility of AMM_AXIOMS App. B.2 (their weight $w \longmapsto \chi_{X/M}$; the former collision with the order width is MOOT — the width is now $\#_\sigma \in \mathcal{I}_{\text{ord}}$), evaluated on per-strike **virtual reserves** in the sense of their App. B.3 ($\alpha, \beta \longmapsto \Delta Q_M^L(i_K), \Delta Q_X^L(i_K)$).

The display is one member of a parameterized class: the subscript tuple is $(\chi_{X/M}, \epsilon_{X/M})$, the second slot the substitution parameter, with $\epsilon_{X/M} = 0$ the Cobb–Douglas member. Whether $\varphi_{(\chi_{X/M},\epsilon_{X/M})}$ satisfies the Bichuch–Feinstein axioms is **not asserted here** — their B.2 alone satisfies all of them (Table 1), but its composition with B.3 virtual reserves is unverified (a later Proposition). **The domain of the flow is OPEN (PR-REGION):** the region over which ΔQ ranges — signedness of the legs and the admissibility set — is not yet defined; no region symbol is minted pending that ruling.

> **FLAG (open, 2026-08-03): LEG ORIENTATION OF $\chi_{X/M}$.** The display above puts $\chi_{X/M}$ on the ΔQ_M leg; the CES definition below puts it on the Q_X leg (and matches Lean `PhiCES.phiCES`). One of the two must change, and the choice flips the $\chi/(1-\chi) = \lambda^{\eta\Delta_i/2}$ bridge and the reading of `curvIndex_is_rho_zero_slice`. Theorem 1 consumes the ΔQ_M-leg form. AUTHOR DECISION REQUIRED — not resolved by the rename.

BINDING (user, 2026-08-03): ϵ is reserved for ELASTICITIES, always subscripted to differentiate; σ is reserved for VOLATILITIES and VARIANCES and is never an elasticity.

Definition 13 (CES trading-function family). Every trading function in this document is a member of ONE two-parameter CES family — $\chi_{X/M}$ the SHARE axis, $\epsilon_{X/M}$ the SUBSTITUTION axis:

$$\varphi_{(\chi_{X/M},\epsilon_{X/M})}(Q_X, Q_M) \equiv \begin{cases} \left(\chi_{X/M}\, Q_X^{\epsilon_{X/M}} + (1-\chi_{X/M})\, Q_M^{\epsilon_{X/M}}\right)^{1/\epsilon_{X/M}}, & \epsilon_{X/M} \neq 0 \\ Q_X^{\chi_{X/M}}\, Q_M^{1-\chi_{X/M}}, & \epsilon_{X/M} = 0 \end{cases} \qquad \chi_{X/M} \in (0,1)$$

$\epsilon_{X/M} = 0$ is a DEFINED CASE, not an evaluation — $1/\epsilon_{X/M}$ is undefined there, so the Cobb–Douglas branch is supplied by definition and CONTINUITY at $\epsilon_{X/M} = 0$ is a **theorem** (`phiCES_tendsto_phiEps`), not a substitution. Every display in this document sits on the $\epsilon_{X/M} = 0$ slice and is subscripted accordingly; Definition 12 is the $\epsilon_{X/M} = 0$ member evaluated on the per-strike virtual reserves. **ORIENTATION (PR-ORIENT, OPEN):** this display carries $\chi_{X/M}$ on the Q_X leg — the **opposite** of Definition 12's ΔQ_M-leg placement; the FLAG applies to the *pair* and one of the two must eventually change.

Definition 14 (Curvature). The **curvature** κ_φ of a trading function is the price-impact elasticity of its marginal price along the trading curve, normalized against the constant-product member: with the **marginal price** minted as its own object — $p_\varphi \equiv \partial_{Q_X}\varphi \,/\, \partial_{Q_M}\varphi$, the quotient of partials of the trading function (bare p is not used; the subscript p in $\epsilon_{p/X}$ names this object; Lean `CurvatureTwo.margPrice`, subject to the PR-ORIENT argument-order FLAG) — and

$$\epsilon_{p/X} \equiv \frac{d\ln p_\varphi}{d\ln Q_X}\Big|_{\varphi=\text{const}}, \qquad \kappa_\varphi \equiv \frac{|\epsilon_{p/X}|}{|\epsilon_{p/X}| + |\epsilon^0_{p/X}|}$$

where $\epsilon^0_{p/X}$ is the same elasticity for the $\epsilon_{X/M} = 0$ (constant-product) member at the same point. $\epsilon_{p/X}$ is an **observable** of any member of the trading-function class (Definition 12), not a parameter: the second derivative of φ enters through it (the derivative of the marginal price), and the benchmark normalization makes κ_φ scale-free, with the constant-product pool at $\kappa_\varphi = 1/2$. Notation binding: κ_φ names the **genuine** curvature (φ the quote function, never the fee ϕ); the share-asymmetry index $\varsigma_{X/M}$ is NOT a curvature.

(The grid–marginal-price relation is Proposition 10, stated with the portfolio-value machinery in # CONTROL_ OPERATORS.)

Proposition 7 (CES curvature closed form). For the CES family (Definition 13), at the balanced point $|\epsilon_{p/X}| = \dfrac{1-\epsilon_{X/M}}{1-\chi_{X/M}}$, and

$$\kappa_\varphi(\epsilon_{X/M}) = \frac{1-\epsilon_{X/M}}{2-\epsilon_{X/M}} \in [0,1), \qquad \epsilon_{X/M}(\kappa_\varphi) = \frac{1-2\kappa_\varphi}{1-\kappa_\varphi}$$

— the $\chi_{X/M}$-dependence cancels identically: κ_φ is a function of the SUBSTITUTION axis alone (equivalently $\kappa_\varphi = 1/(1+\bar{\epsilon}_{X/M})$). The inverse is the DESIGN DIAL — choose a target curvature, read off the substitution exponent. *Status:* **OPEN in-tree** — Aristotle target: (i) the elasticity computation $|\epsilon_{p/X}| = (1-\epsilon_{X/M})/(1-\chi_{X/M})$, (ii) the normalization identity.

Theorem 8 (Properties of the closed form). The closed form is zero exactly at the linear member ($\epsilon_{X/M} = 1$), strictly positive below it, strictly decreasing in $\epsilon_{X/M}$, with range $[0,1)$; $\chi_{X/M}$ and Δ_i do NOT enter it; both round trips with the inverse hold.

Refutation note (what curvature is NOT). The Gaussian curvature of φ's graph is **identically zero** for every member — 1-homogeneity forces $\operatorname{Hess}\varphi \cdot (Q_X, Q_M)^\top = 0$, so $\det \operatorname{Hess} \equiv 0$ and the graph is a ruled surface; the Gaussian reading cannot distinguish linear from Leontief. The un-normalized planar curvature of the trading curve is scale-dependent ($(1-\epsilon_{X/M})/(\sqrt{2}\,t)$ at the symmetric point $Q_X = Q_M = t$, $\chi_{X/M} = 1/2$) and cannot equal a constant. The normalization of Definition 14 is what makes Proposition 7 well-posed. *Status:* unformalized (cheap machine-check; rider on the Proposition 7 Aristotle bundle).

Definition 15 (Share asymmetry). The **share asymmetry** (grid tilt) of the trading-function family is

$$\varsigma_{X/M} \equiv 1 - \Big(\frac{1-\chi_{X/M}}{\chi_{X/M}}\Big)^{\Delta_i}$$

zero exactly at $\chi_{X/M} = 1/2$. It is a **derived observable** — a function of the registered parameters $\chi_{X/M} \in \Theta_\varphi$ and $\Delta_i \in \Theta_p$, adding no degree of freedom; hence no registry entry.

Theorem 9 ($\varsigma_{X/M}$ is not a curvature). At equal shares $\varsigma_{X/M}$ vanishes both at the constant-product member and at the linear member — although the former has $\kappa_\varphi = 1/2 > 0$ (Proposition 7) and only the latter is flat. So $\varsigma_{X/M}$ measures SHARE asymmetry (the grid-price tilt), not curvature across the substitution axis. Blocks E1–E7 are theorems about $\varsigma_{X/M}$: their mathematics stands; their reading is about SHARE, not curvature.

Rule 5 (Current trading curve). The protocol pins the balanced Cobb–Douglas member:

$$\chi_{X/M} \leftarrow \tfrac{1}{2}, \quad \epsilon_{X/M} \leftarrow 0 : \qquad \varphi_{(1/2,0)}(i_K; \Delta Q, L) \;=\; (\Delta Q^L_M(i_K) + \Delta Q_M)^{1/2} \cdot (\Delta Q^L_X(i_K) \;+\; \Delta Q_X)^{1/2}$$

The leg reading — $\chi_{X/M}$ as the ΔQ_M-leg exponent (the $1/p_{(\eta,\Delta_i)}$ leg), that leg's share of pool value — is Definition 12's orientation. At $\chi_{X/M} = 1/2$ the two orientations of the OPEN FLAG coincide, so the current case is orientation-blind — which is why the Definition 12 / Definition 13 contradiction stayed invisible in practice.

Theorem 10 (The bridge $\chi_{X/M} \leftrightarrow \eta \leftrightarrow \varsigma_{X/M}$). The weight ratio IS the per-TICK square-root-price step — so $\chi_{X/M}$ is an OBSERVABLE of the grid already defined, not a new primitive:

$$\frac{\chi_{X/M}}{1-\chi_{X/M}} \;=\; \frac{p_{(\eta,\Delta_i)}(i+1)}{p_{(\eta,\Delta_i)}(i)} \;=\; \lambda^{\eta\,\Delta_i/2}$$

Both directions, both round trips; and the share asymmetry (Definition 15) factors through the share, the per-SPACING step being the per-TICK step raised to Δ_i — with Λ the logistic, $\Lambda(z) = 1/(1+e^{-z})$ (the same Λ the fee schedule uses below):

$$\chi_{X/M}(\eta) \;=\; \Lambda\Big(\frac{\eta\,\Delta_i\,\ln\lambda}{2}\Big) \in (0,1)\;\;\forall\eta, \qquad \eta(\chi_{X/M}) \;=\; \frac{2}{\Delta_i\,\ln\lambda}\,\ln\frac{\chi_{X/M}}{1-\chi_{X/M}}$$
$$\varsigma_{X/M}(\eta,\Delta_i) \;=\; 1-\Big(\frac{1-\chi_{X/M}}{\chi_{X/M}}\Big)^{\Delta_i}, \qquad \chi_{X/M}(\varsigma_{X/M}) \;=\; \frac{1}{1+(1-\varsigma_{X/M})^{1/\Delta_i}}$$

Theorem 11 (Domain coincidence). Three conditions stated independently, in different blocks, are one:

$$\begin{aligned} 0 < \eta\,\Delta_i \quad &\Longleftrightarrow \quad \chi_{X/M} > \tfrac{1}{2} \quad \Longleftrightarrow \quad \varsigma_{X/M} \in (0,1) \\ \eta = 0 \quad &\Longleftrightarrow \quad \chi_{X/M} = \tfrac{1}{2} \quad \Longleftrightarrow \quad \varsigma_{X/M} = 0 \end{aligned}$$

The first line is exactly the hypothesis Theorem 5's `deltaQM_nonneg` requires — an analytic guard that IS the economic condition "the pool is asset-heavy in value"; the second says flat grid = symmetric pool = zero tilt.

Theorem 12 (Share-asymmetry monotonicity; the antitone reading is REFUTED). $\varsigma_{X/M}$ is strictly INCREASING in $\chi_{X/M}$ on $(0,1)$, vanishing exactly at $\chi_{X/M} = \frac{1}{2}$. The opposite reading — a larger asset share means LESS tilt — is FALSE:

$$\Delta_i = 1: \qquad \chi_{X/M} = \tfrac{1}{4} < \tfrac{3}{4} \quad \text{but} \quad \varsigma_{X/M}\big(\tfrac{1}{4}\big) < \varsigma_{X/M}\big(\tfrac{3}{4}\big)$$

(raising $\chi_{X/M}$ shrinks $(1-\chi_{X/M})/\chi_{X/M}$, hence RAISES $1-(\cdot)^{\Delta_i}$.)

CONSEQUENCE FOR E8(6): the factor-share reading was recorded UNAVAILABLE because $\eta^\star \approx 458/\Delta_i^2$ cannot be a Cobb–Douglas share. It never had to be — the share is $\chi_{X/M}(\eta^\star) \in (0,1)$ for EVERY η (Theorem 10), so the identification is reachable through $\chi_{X/M}$, not through η directly. *Carriers:* `etaStar_tilde_mem_Ioo`, `curvIndex_etaStar_via_tilde`. *(E-block cross-note; converts when the E-blocks are swept.)*

Provenance: `EtaTilde` 23/23 axiom-clean, project `67b1c841` (doc $\chi_{X/M} \leftrightarrow$ Lean `etaTilde`, the Lean name fixed by the bundle and never hand-edited); `PhiCES` 12/12 axiom-clean, project `cd3558f7`. Carriers not yet attached to a numbered statement: `phiCES_agreement_point` (evaluation form, scope declared in-file); CONDITIONAL, NOT an identification: `phiCES_rho_vs_pi_eta_trader` gives $1/(1-\epsilon_{X/M}) = 1/(1-\eta) \iff \epsilon_{X/M} = \eta$ away from the poles for `exp/CESLongVolPayoff`'s η, and its docstring states outright that this does NOT identify the payoff parameter with the trading-function parameter — E8(6) untouched.

FEE_ALGEBRA

Rule 6 (Per-leg fee split). Each incoming trade leg is split by its **leg fee** $\phi_M, \phi_X \in (0,1)$ (the M9 leg fees — fee *variables*, produced by the schedule below, not members of Θ_ϕ): the trading function is quoted on the net flow, and the fee fraction accrues to the per-strike reserves —

$$\begin{aligned} \Delta Q_M &= (1-\phi_M)\,\Delta Q_M \;+\; \phi_M\,\Delta Q_M \\ \Delta Q_X &= (1-\phi_X)\,\Delta Q_X \;+\; \phi_X\,\Delta Q_X \end{aligned}$$

$$\begin{aligned} \Delta Q^L_M(i_K) &\leftarrow \Delta Q^L_M(i_K) \;+\; \phi_M\,\Delta Q_M \\ \Delta Q^L_X(i_K) &\leftarrow \Delta Q^L_X(i_K) \;+\; \phi_X\,\Delta Q_X \end{aligned}$$

The first display is an identity (the decomposition); the Rule is the second — the **accrual assignment**: fees deposit into Definition 9's endowments at the strike where they are charged.

Definition 17 (Fee composition). The **fee-composition law** on the carrier $[0,1)$ is

$$\phi_M \otimes_\phi \phi_X \;\equiv\; 1-(1-\phi_M)(1-\phi_X), \qquad \mathcal{G}_\phi \;\equiv\; \big([0,1],\, \otimes_\phi,\, 0\big)$$

$\mathcal{G}_\phi$ is an **Abelian monoid** — identity $\phi = 0$, no inverses (a charged fee cannot be un-charged) — **not a group, and $\otimes_\phi$ is a composition law, not an inner product** (correcting the note's original wording). The monoid axioms are machine-proved — Theorem 14 (`probOr_comm`, `probOr_assoc`, `probOr_zero`, closure `probOr_mem_Icc`). The carrier is $[0,1]$ as proved; the boundary $\phi = 1$ (full confiscation) is admitted by the algebra and excluded economically by Rule 6's domain $\phi_\bullet \in (0,1)$.

Rule 7 (Trader-paid fee). The protocol composes the leg fees into the trader-paid fee by the monoid law:

$$\phi \;\leftarrow\; \phi_M \otimes_\phi \phi_X$$

This is the [M9] **DECIDED** entry — enacted as **Rule 12**, with its algebra as **Theorem 20** (`TauMevAlgebra` carriers), inside the MEV subsection of # CONTROL_OPERATORS, where hazards are introduced.

Design menu. Row 1 is the **DECIDED** structure — Rules 6–7 enact it; the remaining rows are alternative composition structures considered and **not adopted** (candidate Rules never enacted):

Economic process	Operator	Structure
Sequential fee charging	$(1-(1-\phi_M)(1-\phi_X))$	Abelian monoid
Strongest policy wins	(max)	Join semilattice
Cheapest route wins	(min)	Meet semilattice
Liquidity aggregation	Weighted average	Convex algebra
Feature flags	OR	Monoid
Permission intersection	AND	Monoid
Bit toggling	XOR	Abelian group
Cyclic governance states	Addition mod (N)	Finite abelian group

Economic process	Operator	Structure

Definition 18 (Dynamic fee schedule). The fee *level* is produced by the volatility-and-utilization schedule — the sum-of-sigmoids dynamic fee of ALGEBRA (their eq. (4)–(5); α scales, γ steepens, β centers), gated by utilization:

$$\phi\,(\sigma\,(i(t));t)\;\equiv\bar{\phi}\;+\;\Big(\sum_j\frac{\alpha_j}{1+\exp(\gamma_j\,(\beta_j-\sigma(i(t))))}\Big)\cdot\frac{\alpha_R}{1\;+\;\exp(\gamma_R\,(\beta_R-\frac{\varphi_{(1/2,0)}\,(i_K;\Delta Q,0;t)}{\varphi_{(1/2,0)}\,(i_K;0,L;t)}))}$$

The gate's argument is the **utilization ratio** — the trading function evaluated on flow alone over its evaluation on the endowments alone (Theorem 1's u is the gate's value). The parameters $\Theta_\phi=\{\gamma,\bar{\phi},\beta,\alpha\}$ are Protocol Parameters (see **PROTOCOL_PARAMETERS (Θ_ϕ)**); the R-suffixed trio $(\alpha_R,\beta_R,\gamma_R)$ enters as members of the $\alpha/\beta/\gamma$ families. **Signature note:** the t argument extends Definition 12's $(i_K;\Delta Q,L)$ signature — the time dependence enters through the flow and endowments at t; flagged, not silently repaired.

Theorem 1 (Fee Envelope). Writing $u=\alpha_R\Big/\Big(1+\exp\Big(\gamma_R\Big(\beta_R-\frac{\varphi_{(1/2,0)}(i_K;\Delta Q,0;t)}{\varphi_{(1/2,0)}(i_K;0,L;t)}\Big)\Big)\Big)$ — the utilization gate of Definition 18:

$$0\le u\le\alpha_R,\qquad\bar{\phi}\;\le\;\phi(\sigma)\;\le\;\bar{\phi}+\Big(\sum_j\alpha_j\Big)u,\qquad\sigma\longmapsto\phi(\sigma)\text{ monotone }(\gamma_j>0,\alpha_j\ge0,u\ge0)$$

The single-term case is the sigmoid fee schedule with steepness $s_f\;=\;1/\gamma_0$, where Λ is the logistic $\Lambda(z)=1/(1+e^{-z})$:

$$\bar{\phi}+\alpha_0\,\Lambda(\gamma_0(\sigma-\beta_0))\;=\;f\big(\sigma;\,f_{\min}=\bar{\phi},\,f_{\max}=\bar{\phi}+\alpha_0,\,\bar{\sigma}_f=\beta_0,\,s_f=\gamma_0^{-1}\big)$$

`VolInstrument.sigmoidR_mem`, `multiFee_bounds`, `multiFee_monotone`, `multiFee_single_bridge`.

ECONOMIC CONTENT OF THEOREM 1. The floor $\bar{\phi}$ is unconditional — LPs take a base fee at every volatility, so the schedule never degenerates to free execution. The ceiling is NOT a constant: it is $\bar{\phi}+(\sum_j\alpha_j)\,u$, GATED by the utilization factor $u\in[0,\alpha_R]$, so a pool nobody is trading against cannot levy the volatility surcharge at all — at $u\;=\;0$ the band collapses to the floor, and the surcharge is earned only where flow exists to earn it on. Monotonicity in σ is what makes the schedule a genuine VOLATILITY SURCHARGE rather than an arbitrary function of state: higher realized volatility always costs the trader weakly more. That is the property the FLAIR identification consumes — $\Theta_{\lambda_{\text{FLAIR}}}=\{\bar{\phi},\alpha,u\}$ is a LEVEL block precisely because the band's two edges are the level parameters, while (β_j,γ_j) only place the transition inside the band (G3).

Rule 2 (Streamia). Assign the per-time-step payoff variation to the trading fee — the *streamia* of the Panoptic whitepaper:

$$\phi\,(\sigma\,(i(t));t)\;\overset{\text{streamia}}{\longleftarrow}\;\theta\Big(\,p_{(\eta,\Delta_i)}\,(i;t)\,,p_{(\eta,\Delta_i)}\,(i_K),\sigma\,(i(t))\Big)$$

$\overset{\text{streamia}}{\longleftarrow}$ denotes this Rule throughout.

Definition 3 (Streaming premium). The **streaming premium** over N steps of length Δt is the accumulated streamia $\sum_{j<N}\theta_j\,\Delta t$ — the seller's fee income under Rule 2. It replicates the option's time decay, which is exactly why the perpetual instrument needs no expiry: the decay that a dated option pays out at T is instead streamed continuously.

PROTOCOL_INPUTS

Every **Protocol Input** is a quantity supplied by the USER, per order, at interaction time — the third registry class. The classifying test across the three registries is *who sets it, and when*: a Protocol Constant ($\mathcal{C}_p$) is fixed by the design once and forever; a Protocol Parameter ($\Theta_{\bullet}$) is set by the protocol, uniformly for all users; a Protocol Input is chosen by the user for each interaction. Input entries carry a *Carrier* line — inputs are calldata, and the on-chain field is part of their definition. The input set of the **vol order** (retiring the former Θ_{ord} symbol: inputs are not parameters, so the index letter changes):

$$\mathcal{I}_{\text{ord}} \;=\; \{\sigma_K^2,\; \#_\sigma,\; s_\upsilon\}$$

(strike, width, skew) pins only the scale-free leg shape $\ell\,(\xi, \iota; i_K)$.

Rule 8 (Target-vega completion — DECIDED, 2026-07-30). The order is completed with the target vega:

$$\mathcal{I}_{\text{ord}} \;\leftarrow\; \mathcal{I}_{\text{ord}} \cup \{\Delta Q_\upsilon^\star\}$$

Protocol Input ($\mathcal{I}_{\mathbf{ord}} = \{\sigma_K^2, \#_\sigma, s_\upsilon, \Delta Q_\upsilon^\star\}$ — the vol order).

- σ_K^2 — the **strike variance**. *Domain:* a variance level (Convention 2: $\sigma^2(i_K) \equiv \sigma_K^2$); on-chain packing per `VolOrderValidationLib`. *Purpose:* the strike of the volatility option (Definition 1). *Economic meaning:* the variance level above which the option pays. *Carrier:* `create_order(strike, ...)`.
- $\#_\sigma$ — the **width** (symbol per user ruling 2026-08-04; formerly w, retired — the AMM_AXIOMS w-collision noted at Definition 12 is thereby MOOT). *Domain:* the packed order field (validation predicate in `VolOrderValidationLib`). *Purpose:* with s_υ, pins the scale-free leg shape $\ell(\xi, \iota; i_K)$. *Economic meaning:* the strike-band width of the replication ladder. *Carrier:* `create_order(..., width, ...)`.
- s_υ — the **skew** (symbol per user ruling 2026-08-04; formerly bare s, retired — it collided with the fee-schedule steepness s_f). *Domain:* the packed order field (validation predicate in `VolOrderValidationLib`). *Purpose:* with $\#_\sigma$, pins the leg shape. *Economic meaning:* the asymmetry of the ladder around the strike. *Carrier:* `create_order(..., skew, ...)`.
- $\Delta Q_\upsilon^\star$ — the **target vega** (enters by Rule 8). *Domain:* `u96`, RAW LIQUIDITY units — $\Delta Q_\upsilon^\star$ carries the dimension of the replication carrier L (the DECIDED dimension ruling below). *Purpose:* sizes the ladder and induces the implied maturity (Theorem 13, # PAYOFF). *Economic meaning:* the vega notional the user targets — the one sizing decision the order stores. *Carrier:* `targetVega : u96` at bits 152..247 of the packed `VolOrder` word (plank `feat/plank`); `targetVega` $= \Delta Q_\upsilon^\star$ exactly; emitted by `VolOrderCreated(orderId, strike, width, skew, targetVega)`; fits-packed predicate in `VolOrderValidationLib`.

Convention 3 (Vega dimension — stored target vs lens readout; DECIDED 2026-07-30). $\Delta Q_\upsilon^\star$ carries the dimension of the REPLICATION CARRIER — liquidity L, the quantity of the priced vol asset, per Rule 4's ledger

$$\pi^\sigma \;=\; \sum_{i_K} L(i_K)\,\mathbb{I}_{\text{long|short}}$$

The quotient $\Delta Q_\upsilon \equiv \Delta\pi^\sigma / \Delta(\sigma^2 - \sigma_K^2)^+$ (collateral per vol unit, Definition 1) is the LENS READOUT — computed from a position through the Q_M^L range conversion (Definition 10), never stored. One instrument, two views: the $\mathcal{I}_{\mathrm{ord}}$ entry names the **stored quantity**, Definition 1's quotient names the **measured sensitivity**.

Rule 9 (Sizing — quantity-exact, no price in the map). The mint sizes the ladder from the target vega alone:

$$L(i_K) \leftarrow \Delta Q_v^\star\, \ell\,(\xi^\star, \iota; i_K), \qquad \sum_{i_K} L(i_K) = \Delta Q_v^\star \;\; \big(\textstyle\sum_{i_K} \ell = 1,\ \text{Theorem 2}\big)$$

$p_{\mathrm{vol}}, p_{\mathrm{risk}}$ enter at the ISSUANCE/ADMISSIBILITY layer (shares, deleverage — Rule 10), never the sizing map; the mint's collateral requirement is the actual replication cost, slippage-bounded.

Proposition 8 (The lens obligation). Delivered quantity recovers the stored target, one-sided under per-leg floor rounding:

$$\sum_{i_K} L(i_K) \le \Delta Q_v^\star$$

Status: **OPEN in-tree** — the per-leg floor rounding of Rule 9's map has no Lean carrier (the induced-ladder floor-maximal construction is plank-side). The *adjacent* rounding conservativity that IS proved is the funded-cap side of Rule 10: `dQvFunded_roundDown`, `roundDown_preserves_invariant` — related, not carriers of this statement.

Rule 10 (Collateral channel and auto-deleverage; DECIDED 2026-07-30). The contract holds $\Delta Q_v^\star$ fixed, so all adaptation lands on collateral. The live backing requirement, and the (division-free) admissibility condition:

$$\Delta M_{\mathrm{req}}(t) \leftarrow \Delta Q_v^\star \cdot p_{\mathrm{vol}}(\bar\sigma; t), \qquad \Delta Q_v \cdot p_{\mathrm{risk}}(t) \le Q_M$$

On violation the position is NOT hard-liquidated: the enforced exposure contracts to the funded level,

$$\Delta Q_v(t) \leftarrow \min\Big(\Delta Q_v^\star,\ \frac{Q_M(t)}{p_{\mathrm{risk}}(t)}\Big) \quad \text{(floor)}, \qquad T^\star(t) \leftarrow 2\,\frac{\Delta Q_v(t)}{N_\sigma}$$

so the implied maturity CONTRACTS continuously with the funded exposure instead of truncating; a top-up restores both. Liquidation is the degenerate case $Q_M \to 0$, where the realized life $[t_{\mathrm{mint}}, t_{\mathrm{liq}}]$ is the maturity the position actually had. **FLAG (define-before-use, OPEN):** p_{vol}, p_{risk}, and the reference volatility $\bar\sigma$ are consumed here but not yet defined in the converted region — they are plank-side price feeds (the `priceOfRisk` entry point); their formal definitions are owed before this Rule's symbols close.

$$\forall x,\ 0 \le x \le \Delta Q_v^\star \wedge x\, p_{\mathrm{risk}} \le Q_M \;\Rightarrow\; x \le \Delta Q_v(t), \qquad \Delta Q_v(t)\, p_{\mathrm{risk}} \le Q_M \;\; \text{(on violation)}$$

`dQvFunded_maximal`; `dQvFunded_admissible(_iff_mul)`, `_mul_le_of_violation`, `_eq_of_no_violation`; $T^\star(t) \uparrow Q_M, \downarrow p_{\mathrm{risk}}$: `tStarFunded_mono_QM`, `_antitone_prisk`; $Q_M \ge \Delta Q_v^\star p_{\mathrm{risk}} \Rightarrow T^\star(t) = T^\star$: `_eq_tStar_of_topup`; $Q_M = 0 \Rightarrow T^\star(t) = 0$: `dQvFunded_zero_QM`; floor rounding conservative (min-monotone).

Rule 11 (Recalibration law; DECIDED 2026-07-30: multiplicative). The joint evolution of the implied maturity under the collateral channel AND realized variance $\sigma_R^2(t)$ accruing against the strike:

$$T^\star_{\mathrm{joint}}(t) \leftarrow T^\star(t) \cdot \Big(1 - \frac{\sigma_R^2(t)}{\sigma_K^2}\Big)^+ = \underbrace{\frac{2\,\Delta Q_v^\star}{N_\sigma}}_{T^\star} \cdot \underbrace{\frac{\min\big(\Delta Q_v^\star,\ Q_M/p_{\mathrm{risk}}\big)}{\Delta Q_v^\star}}_{\text{funding factor}} \cdot \underbrace{\Big(1 - \frac{\sigma_R^2}{\sigma_K^2}\Big)^+}_{\text{budget factor}}$$

The arrow is the Rule (the enacted law); the second equality is the factorization identity. *Rationale (recorded):* $\upsilon = T/2 \Rightarrow \sigma^2$-budget $\propto T$ (bijection preserved); $T^\star_{\text{joint}} = T^\star \cdot f_{\text{fund}} \cdot f_{\text{budget}}$ (monotonicities chain); burn rate constant (no cliff).

> NOTE (cascade, recorded): $\Delta Q^\star_\upsilon$ on-chain lands on the PAIR (PanopticTokenId, positionSize) — the tokenId is scale-free (strikes, widths, per-leg optionRatio); positionSize is an SFPM mint argument. The ratio-vs-size split of $\ell(\xi^\star, \iota; i_K)$ across the pair is the task-#14 sizing decision. Spec: `.planning/vol-order-v2-target-vega-SPEC.md`.

CONTROL_OPERATORS

These are the instruments mapping **behavior objectives** to **protocol parameters** — all of $\Theta_\bullet$, not only the fee schedule Θ_ϕ.

Convention 4 (Hazard rate vs incidence operator). The control operators come in two types, distinguished by the glyph: plain $\lambda_\bullet$ names a **hazard rate** — an arrival intensity (probability per unit time) of a behavior (arb toxicity, LP competition); tilde $\tilde{\lambda}_\bullet$ names an **incidence operator** — a re-routing of already-arriving flow that leaves the total invariant. A hazard *adds* under composition ($\bigoplus$, Definition 19); an incidence operator *applies* — it changes who bears the flow, not how much arrives. The distinction is proved, not stylistic: the JIT operator leaves `mevTotal` invariant while FLAIR falls and the toxicity ratio rises (`JitLiquidity` carriers) — it cannot be a hazard.

Definition 19 (Hazard aggregation). The aggregate hazard is the composition of the behavior hazards, and $\bigoplus$ on hazard rates **is addition** (the hazard-coordinate image of $\otimes_\phi$ — Theorem 14):

$$
\begin{aligned}
\lambda &\equiv \bigoplus_{i=1}^{n} \lambda_i \qquad i \in \{\text{lp-competition (FLAIR)}, \text{arb toxicity}, \text{TBD}, \cdots\} \\
\lambda &\equiv \lambda_M + \lambda_X
\end{aligned}
$$

MEV is struck from the index set (correcting the note's original listing): λ_{ARB} — already a member — is λ_{MEV}'s SUMMAND (Definition 23), so listing MEV double-counts; and the MEV tax enters the trader-paid fee through the $\otimes_\phi$ monoid (Rule 12), never through $\bigoplus$. The second line is the per-leg split, mirroring the leg fees of Rule 6.

Theorem 14 (Fee monoid and hazard exactness). $\mathcal{G}_\phi = \big([0,1], \otimes_\phi, 0\big)$ is an Abelian monoid — commutative, associative, identity 0, closed on $[0,1]$, monotone — and the hazard correspondence is **exact** under $\phi = 1 - e^{-\lambda}$:

$$
\big(1 - e^{-\lambda_M}\big) \otimes_\phi \big(1 - e^{-\lambda_X}\big) = 1 - e^{-(\lambda_M + \lambda_X)} \iff \lambda \equiv \lambda_M + \lambda_X
$$

Definition 19's $\bigoplus$-is-addition is exactly this exactness: fee composition and hazard addition are the same law in two coordinate systems.

Definition 24 (Linear pool value). The **linear pool value** is the pool's holdings marked at spot — the money-units valuation with no curvature adjustment:

$$
\pi^{\text{linear}}(t) \equiv p_{(\eta,\Delta_i)}(i(t))\, Q^L_X\Big(\textstyle\sum_j^{\#\text{LP}} L_j(i(t);\cdot)\Big) + Q^L_M\Big(\textstyle\sum_j^{\#\text{LP}} L_j(i(t);\cdot)\Big)
$$

(Symbol per user rulings 2026-08-04: values are π-objects and this valuation is linear; the former ad-hoc D_t is retired — D reads as debt.)

Recall (the marginal price). $p_\varphi \equiv \partial_{Q_X}\varphi \,/\, \partial_{Q_M}\varphi$ — the quotient of partials of the trading function, minted at Definition 14; its relation to the grid is the next statement, placed here because Definitions 25–26 consume both objects (user ruling 2026-08-04).

Proposition 10 (Grid–marginal-price relation). The grid map and the marginal price are DISTINCT objects — the identification $p_{(\eta,\Delta_i)} \equiv p_\varphi$ is NOT admissible. At Definition 9's reserves, for the $\chi_{X/M} = 1/2$ member,

$$p_\varphi(i_K) \;=\; \frac{\Delta Q^L_M(i_K)}{\Delta Q^L_X(i_K)} \;=\; \frac{1}{p_{(\eta,\Delta_i)}(i_K)\; p_{(\eta,\Delta_i)}(i_K + \Delta_i)}$$

— the grid enters the marginal price as the INVERSE PRODUCT of adjacent grid values: the $\sqrt{\text{price}}$-vs-price gap Theorem 4 already flags (`priceGrid_eq_tickPrice_sq`), plus the leg orientation. *Status:* elementary algebra from Theorem 5's reciprocal form; **unproved in-tree** (cheap Aristotle rider on the Proposition 7 bundle).

Definition 25 (Portfolio value function). With $(Q^L_X(p_\varphi), Q^L_M(p_\varphi))$ the point of the trading curve $\varphi_{(\chi_{X/M},\epsilon_{X/M})} = \text{const}$ at which the marginal price p_φ (Definition 14) attains a given value, the **portfolio value function** is the on-curve valuation of the RESERVES — the L-superscripted quantities (Definition 10): bare Q_X, Q_M remain the trading-side arguments of Definition 13, while the reserves derived from liquidity are what the pool holds and what is valued here —

$$\pi^\varphi(p_\varphi) \;\equiv\; p_\varphi\, Q^L_X(p_\varphi) \;+\; Q^L_M(p_\varphi)$$

— the portfolio value function of CFMM_GEOMETRY, the conic dual of the trading function (their equivalence theorem); concave and nondecreasing in p_φ. **Relation to Definition 24:** π^{linear} marks FIXED holdings at spot; π^φ moves holdings ALONG the curve — at the current price the two coincide, away from it π^φ falls below the fixed-holdings line, and that concavity gap is what LVR prices. *Status:* **UNFORMALIZED** — no Lean carrier (`exp/CESLongVolPayoff.pi_eta_trader` is the trader-side Bregman object, distinct).

Definition 26 (LVR rate). The loss-versus-rebalancing rate is a PAYOFF-shaped object — a loss — hence the π glyph (user ruling 2026-08-04): π^{LVR}, carrying NO time argument (it is a state function of the current tick); the time-argument form $\pi^{\text{LVR}}(t)$ is the per-block realized loss below, so no bar normalization is needed. Per MMR, with the second derivative well-defined on Definition 25's object — and the evaluation point CORRECTED (user-exposed 2026-08-04) to the current MARGINAL price $p_\varphi(i(t))$, not the grid value, the two differing by Proposition 10's relation:

$$\pi^{\text{LVR}} \;\equiv\; \frac{\sigma^2(i(t))\, p_\varphi^2}{2} \left|\frac{d^2\pi^\varphi}{dp_\varphi^2}\right| \Big|_{p_\varphi = p_\varphi(i(t))} \qquad \left(\text{CPMM: } \pi^{\text{LVR}} = \frac{\sigma^2}{8}\,\pi^\varphi\right)$$

Discretization frame (t-indexed, shared by FLAIR and MEV; the Lean carriers keep their `w_t/D_t/a_t` names — standing doc-glyph/Lean-name split). Time is stepped by the cadence Δt; per step t:

- the per-step traded VOLUME in LIQUIDITY UNITS is $\varphi_{(1/2,0)}\big(i(t);\, \Delta Q(t),\, 0\big) \;=\; \sqrt{\Delta Q_M(t)\,\Delta Q_X(t)} \;\geq\; 0$ — Definition 18's zero-liquidity convention: the symmetric geometric mean of the two legs, neither money nor asset alone, commensurable with L and $\Delta Q^\star_v$; NO alias symbol is minted for it (the former $\Delta Q_\cdot(t)$ is retired — user ruling 2026-08-04);
- $\pi^{\text{linear}}(t) > 0$ — the per-step capital in MONEY units (Definition 24), serving the MEV/LVR side, where LVR is intrinsically a money rate;
- $\pi^{\text{LVR}}(t) \;\equiv\; \pi^{\text{LVR}} \cdot \Delta t \;\geq\; 0$ — the per-block arb-opportunity weight: the rate (Definition 26, no time argument) over one block; the time argument itself marks the per-block object, so no bar normalization is needed (user ruling 2026-08-04);

- $\nu_t \equiv \varphi_{(1/2,0)}\big(i(t);\, \Delta Q(t),\, 0\big) \,/\, \varphi_{(1/2,0)}\big(i(t);\, 0,\, L\big)$ — the PER-STEP UTILIZATION RATIO, exactly Definition 18's gate argument: FLAIR's capital-normalized flow, dimensionless with no numéraire choice.

Coordinates (user ruling (i), 2026-08-04): FLAIR runs in UTILIZATION coordinates (ν_t); MEV runs in MONEY coordinates ($\pi^{\mathrm{LVR}}(t)/\pi^{\mathrm{linear}}(t)$). No other t-indexed symbols are introduced in this section.

Definition 30 (HODL value). The inception basket marked at the current price — the tangent line to π^{φ} at $p_{\varphi}(t_0)$:

$$\pi^{\mathrm{HODL}}(t) \;\equiv\; p_{(\eta,\Delta_i)}(i(t))\, Q_X^L(t_0) \;+\; Q_M^L(t_0)$$

$\pi^{\mathrm{HODL}}(t_0) \;=\; \pi^{\mathrm{linear}}(t_0)$, and thereafter $\pi^{\mathrm{HODL}}(t) \;\geq\; \pi^{\varphi}(p_{\varphi}(t)) \;=\; \pi^{\mathrm{linear}}(t)$ (tangent above the concave curve) — the gap's expected rate is π^{LVR} (Definition 26).

Definition 31 (Returns). Gross returns in the form of [DUFFIE] (D. Duffie, *Dynamic Asset Pricing Theory*, 3rd ed., Princeton UP, 2001 — the book is copyrighted and not vendored; author-hosted companions: survey, revisions): payoff over INCEPTION capital — the denominator must be t_0 (a contemporaneous denominator makes $R^{\varphi} \equiv 1$ by tangency):

$$R^{\varphi}(t) \;\equiv\; \frac{\pi^{\varphi}(p_{\varphi}(t))}{\pi^{\mathrm{linear}}(t_0)}, \qquad R^{\mathrm{HODL}}(t) \;\equiv\; \frac{\pi^{\mathrm{HODL}}(t)}{\pi^{\mathrm{linear}}(t_0)}$$

FLAIR

Definition 20 (FLAIR). The **LP-competition hazard** λ_{FLAIR} is the time-integrated fee yield per unit of pooled capital — the FLAIR metric of FLAIR, instantiated on this document's objects:

$$\lambda_{\mathrm{FLAIR}}\,(t) \;\equiv\; \int_{t_0}^{t} \frac{\displaystyle\int_{p_{(\eta,\Delta_i)}\,(i(t))} \phi\,(\sigma\,(i(t)); t)\, d\, p_{(\eta,\Delta_i)}\,(t)}{\varphi_{(1/2,0)}\Big(i(t);\, 0,\, \sum_j^{\#\mathrm{LP}} L_j\,(i(t); \cdot)\Big)}\, dt$$

— numerator: the fee density collected across the price range; denominator: the pool capital in **LIQUIDITY UNITS** — the trading function at zero flow on the aggregate liquidity, per Definition 18's utilization convention (user ruling (i), 2026-08-04: FLAIR is utilization-based; the money-units π^{linear} serves the LVR/MEV and ADL layers). It is a plain-λ hazard (Convention 4): fee income *arrives*; nothing is re-routed. **Two repairs vs the raw note (user-approved 2026-08-04):** the undeclared $p_{(\cdot)}$ contraction is expanded to $p_{(\eta,\Delta_i)}$ (no new shorthand minted), and the denominator's first term was corrected $Q_M^L \to Q_X^L$ (both terms were the money leg) — that money-units form was then SUPERSEDED by the utilization-coordinates restatement above.

Theorem 15 (FLAIR identification and corner solution). The program $\sup_{\Theta_{\lambda_{\mathrm{FLAIR}}}} \lambda_{\mathrm{FLAIR}}$ over a sub-block $\Theta_{\lambda_{\mathrm{FLAIR}}} \subset \Theta_{\phi}$ is **identified and solved**. Discretizing per the frame (ν_t the per-step utilization ratio; Λ the logistic of Theorem 10):

$$\lambda_{\mathrm{FLAIR}} \;=\; \bar{\phi}\, W \;+\; u \sum_j \alpha_j\, W_j, \qquad W = \sum_t \nu_t, \quad W_j = \sum_t \Lambda\big(\gamma_j(\sigma(i(t)) - \beta_j)\big)\, \nu_t, \quad 0 \leq W_j < W$$

$$\Theta_{\lambda_{\mathrm{FLAIR}}} \;=\; \{\bar{\phi},\, \alpha,\, u(\alpha_R)\} : \qquad \lambda_{\mathrm{FLAIR}} \;\leq\; \Big(\bar{\phi}_{\max} + u_{\max} \sum_j \alpha_{j,\max}\Big)\, W$$

attained **bang-bang at the level corner** for any fixed (β, γ); in (β, γ) the bound is approached only as $\beta \to -\infty$ — a saturation boundary, not a maximum: the shape parameters never attain it (strict gap at every finite β). This is the G3 level/shape split: $\Theta_{\lambda_{\text{FLAIR}}}$ is a LEVEL block; (β_j, γ_j) only place the transition.

Caveat (kept): this functional has no demand elasticity — the fee–volume trade-off lives in the optimal-fee layer (`FeeSchedule`, arXiv:2508.08152). *Note (OPEN):* Rule 6 charges fees PER LEG; the discretization applies the composed fee (Rule 7) to volume — the leading-order equivalence of per-leg fee income and composed-fee-on-volume is assumed, unformalized.

MEV

Sources, all vendored: MMR (the anchor — arb profits with fees), MEV_THEORY_I (arXiv 2207.11835, the sandwich channel), OPT_FEES (arXiv 2508.08152, the optimal-fee layer). Angstrom is the implementation reference (batch auction / uniform clearing). Statements below carry their provenance tags [M0]–[M10]; the former standalone M-blocks are consumed by this section (byte-pins on them are invalidated by the move — disclosed).

Convention 5 (Event probabilities) [M0]. Probabilities are written $\mathbb{P}_{\text{event}}$: $\mathbb{P}_{\Delta_{\text{ARB}}}$ = arbitrage-trade probability (the paper's `P_trade`; Lean `MevOptimization.ptrade`), $\mathbb{P}_{L_{\text{JIT}}}$ = JIT-arrival probability (CJZ's π; Lean π`J`).

Notation map [M0]. MMR's fee symbol γ is transcribed as this document's fee ϕ; this document's γ`_j` stays the sigmoid steepness. The paper's Poisson block rate λ is transcribed through its own primitive Δ`t` $\triangleq$ λ`⁻¹`, because this document's λ is the hazard rate (Convention 4). The paper's composite parameter η $\triangleq$ $\gamma\sqrt{(2\lambda)/\sigma}$ is deliberately never named — η is reserved project-wide for the pricing grid (Definition 8). Root-block-rate factor: $\sqrt{2/\Delta t}$ throughout, no composite abbreviation. Fee = ϕ (ceiling $\bar{\phi}$, set Θ_ϕ); the quote function is $\varphi_{(\chi_{X/M}, \epsilon_{X/M})}$ (Definition 13), currently $\varphi_{(1/2,0)}$ (Rule 5); bare φ is NOT used.

Δt: mean interblock time (Angstrom: 1 bundle/block/pair $\Rightarrow$ batch cadence = Δt). $\sigma(i(t))$ enters BOTH the fee and $\mathbb{P}_{\Delta_{\text{ARB}}}$ — always written in full tick-argument form (Convention 2; no σ_t shorthand). The t-indexed symbols $\pi^{\text{linear}}(t)$, $\pi^{\text{LVR}}(t)$, ν_t (and the unaliased traded volume $\varphi_{(1/2,0)}(i(t); \Delta Q(t), 0)$) are the discretization frame at the head of this section (FLAIR in utilization coordinates, MEV in money coordinates — user ruling (i), 2026-08-04).

λ_{ARB} (Definition 22) $\subsetneq$ λ_{MEV} (Definition 23): SUMMAND, not sibling — Definition 19's index set carries one, never both (double-count); λ_{ARB} absorbs the "arb toxicity" entry. The paper's `FEE` $\subsetneq \lambda_{\text{FLAIR}}$ (noise flow excluded there). Standing hypotheses: the paper's Assumption 2 (symmetric driftless mispricing, two-sided fee; non-symmetric variant App. C); Proposition 9 additionally: regularity (13), (15).

Definition 21 (Arbitrage-trade probability) [M1]. Per MMR Thm 1 (§4.1, Assumption 2) — the long-run fraction of blocks with a profitable arb; bonding-function-independent, only the fee enters:

$$\mathbb{P}_{\Delta_{\text{ARB}}}\big(\phi(\sigma(i(t)); t), \sigma(i(t)), \Delta t\big) \equiv \frac{\sigma(i(t))}{\sigma(i(t)) + \phi(\sigma(i(t)); t)\sqrt{2/\Delta t}}$$

Both slots are instantiated (user comments 2026-08-04): the σ slot at the realized tick volatility $\sigma(i(t))$ (Convention 2), the ϕ slot at the schedule value $\phi(\sigma(i(t)); t)$ (Definition 18). Lean's `ptrade` keeps both slots abstract — Theorem 16's monotonicities and convexity are statements in the abstract ϕ slot.

Theorem 16 (Properties of $\mathbb{P}_{\Delta_{\text{ARB}}}$) [M1]. $\mathbb{P}_{\Delta_{\text{ARB}}} \in (0, 1]$, with $\mathbb{P}_{\Delta_{\text{ARB}}} = 1 \iff \phi = 0$; strictly decreasing AND **strictly convex** in ϕ; increasing in Δt and in σ; $\to 0$ as $\phi \to \infty$. (The strict convexity is what Theorem 19's strict half consumes.)

Proposition 9 (The MMR split) [M2]. At fast-block small-fee leading order ($\approx$ inherited by everything below), the rebalancing loss splits by the trade probability. In this document's π-convention (payoff/value objects; user ruling 2026-08-04): the paper's `ARB` is the arb-extracted payoff π^{ARB}, its `FEE` is the fee-income payoff π^{ϕ} (fee glyph ϕ — DISTINCT from π^{φ}, the trading-function glyph, per the standing ϕ/φ split), and its `LVR` is the loss payoff π^{LVR} (Definition 26):

$$\pi^{\mathrm{ARB}} \approx \pi^{\mathrm{LVR}} \cdot \mathbb{P}_{\Delta_{\mathrm{ARB}}}, \qquad \pi^{\phi} \approx \pi^{\mathrm{LVR}} \cdot (1 - \mathbb{P}_{\Delta_{\mathrm{ARB}}}), \qquad \pi^{\mathrm{ARB}} + \pi^{\phi} \approx \pi^{\mathrm{LVR}}$$

Status: asserted from MMR Thm 3 + eq. (12), Thm 4 — a Proposition, not a Theorem: the in-tree `arb_add_fee_eq_lvr` is a bridge identity only and is never to be cited as MMR Thm 3 formalized.

Definition 22 (The discrete λ_{ARB}) [M3]. The ARB-channel hazard, on the SAME Θ_{ϕ} as FLAIR ($\phi(\sigma) =$ `multiFee`$(n, \gamma, \beta, \alpha, \bar{\phi}, u)$):

$$\lambda_{\mathrm{ARB}}(t) \equiv \sum_{s<t} \mathbb{P}_{\Delta_{\mathrm{ARB}}}\big(\phi(\sigma(i(s))), \sigma(i(s)), \Delta t\big) \, \frac{\pi^{\mathrm{LVR}}(s)}{\pi^{\mathrm{linear}}(s)}$$

The running-time argument mirrors $\lambda_{\mathrm{FLAIR}}(t)$ (Definition 20); s is the step dummy (user comment 2026-08-04).

CPMM instantiation — NOT a new definition: both tiers instantiate Definition 26's object on the CPMM member, per MMR §7.1 (leading order) and Corollary 2 (exact). Two tiers: (i) the LEADING-ORDER per-step weight

$$\pi^{\mathrm{LVR}}(t) = \frac{\sigma^2(i(t))}{8} \, \pi^{\varphi}(t) \, \Delta t$$

(Definition 26's CPMM case — π^{LVR} is a RATE $\Rightarrow \cdot\Delta t$ per block; summand $\propto \Delta t^{3/2}$ = MMR §7.1 per-block scaling; no guard needed); (ii) the EXACT Corollary-2 kernel, restated in this document's objects (π^{ARB}, π^{φ}, Convention 2 — the paper's `ARB/V` form was stale notation):

$$\big(\pi^{\mathrm{ARB}}/\pi^{\varphi}\big)_{\mathrm{exact}} = \frac{\big(\sigma^2(i(t))/8\big) \, \mathbb{P}_{\Delta_{\mathrm{ARB}}} \, e^{\phi/2}}{1 - \sigma^2(i(t)) \, \Delta t/8}$$

— the ONLY object carrying the guard $\sigma^2(i(t)) \, \Delta t < 8$; reuse this symbol downstream.

Theorem 17 (Identification of $\Theta_{\lambda_{\mathrm{ARB}}}$) [M4]. For $\gamma_j > 0$: λ_{ARB} is decreasing in $\bar{\phi}$, α_j, u, increasing in β_j, convex in ϕ; and there is **no affine analogue** of Theorem 15's `flairMulti_affine` — $\mathbb{P}_{\Delta_{\mathrm{ARB}}}$ is non-affine, level/shape do not separate, Theorem 18's bound is a SUM, not scalar × path weight. The identified block:

$$\Theta_{\lambda_{\mathrm{ARB}}} = \{\bar{\phi}, \, \alpha, \, u\}$$

Batch clearing (Definition 23, $\lambda_{\mathrm{sandwich}} = 0) \Rightarrow \Theta_{\lambda_{\mathrm{MEV}}} = \Theta_{\lambda_{\mathrm{ARB}}} = \{\bar{\phi}, \alpha, u\}$.

Theorem 18 (The infimum program on λ_{ARB}) [M5].

$$\lambda_{\mathrm{ARB}}(t) \geq \sum_{s<t} \mathbb{P}_{\Delta_{\mathrm{ARB}}}\Big(\bar{\phi}_{\max} + u_{\max} \textstyle\sum_j \alpha_{\max,j}, \; \sigma(i(s)), \; \Delta t\Big) \frac{\pi^{\mathrm{LVR}}(s)}{\pi^{\mathrm{linear}}(s)}$$

Three attainment statements (the RHS uses the fee CEILING — unreachable at finite shape): (i) fixed shape $\Rightarrow$ the level-block infimum is attained bang-bang at the corner TOP; (ii) the bound is approached only as $\beta_j \to -\infty$, with a STRICT gap at every finite β (a saturation boundary, not a minimum); (iii) on a compact box a minimizer exists, with value strictly above the bound.

Annotation [M6a] (internal reference — deliberately not a numbered statement, user ruling 2026-08-04): over Θ_{ϕ} unconstrained there is NO trade-off — max λ_{FLAIR} and min λ_{ARB} sit at the SAME level corner, saturate along the SAME $\beta_j \to -\infty$, robustly to every linear scalarization; (β, γ_j) are NOT essential.

Carriers: `joint_corner_degeneracy`, `joint_beta_degeneracy`, `joint_scalarization_degeneracy` (`MevJointProgram.lean`). The degeneracy-breaker must come from OUTSIDE Θ_ϕ.

Theorem 19 (Flat-path optimality at constant σ; the σ-varying comparison is REFUTED) [M6b]. Over arbitrary nonnegative fee PATHS $\{\phi_t\}$ — NOT Θ_ϕ schedules — with ν_t per the frame, $W = \sum_t \nu_t > 0$, the FLAIR fee budget $B \equiv \sum_t \phi_t \nu_t$ (the fee income the path is constrained to deliver), aligned measure $\pi^{\mathrm{LVR}}(t) \equiv \varphi_{(1/2,0)}(i(t); \Delta Q(t), 0)$, and constant volatility $\sigma(i(t)) \equiv \sigma(i(t_0))$:

$$\lambda_{\mathrm{ARB}} \;\geq\; W \cdot \mathbb{P}_{\Delta_{\mathrm{ARB}}}\!\left(\frac{B}{W},\, \sigma(i(t_0)),\, \Delta t\right), \qquad \text{equality} \iff \phi_t \text{ constant on } \{t \,:\, \nu_t > 0\}$$

B/W is the **budget-mean fee** — the flat fee delivering the same FLAIR income as the path (`flair_budget_pins_mean_fee`) — the flat path minimizes λ_{ARB} at equal FLAIR income; non-constant on $\{\nu_t > 0\}$ is strictly worse (the strict half consumes Theorem 16's strict convexity). The alignment $\pi^{\mathrm{LVR}}(t) \equiv \varphi_{(1/2,0)}(i(t); \Delta Q(t), 0)$ is STRONG (traded volume $\propto$ LVR path block-by-block — and CROSS-COORDINATE under ruling (i): a liquidity-units path proportional to a money-units path); without it Jensen is inapplicable and the conclusion can reverse. **REFUTED for σ-varying schedules** (`mev_ge_flat_under_flair_budget_false`): $\exists\, \phi(\cdot) \geq 0$ with $\lambda^{\mathrm{flat}}_{\mathrm{ARB}} > \lambda^{\phi}_{\mathrm{ARB}}$ at equal FLAIR income — witness $T{=}2,\ \Delta t{=}2,\ B{=}2,\ \sigma = (1, 10)$, fees $(2, 0)$: $\frac{31}{22} > \frac{4}{3}$ (σ-varying $\Rightarrow$ different convex summands, Jensen inapplicable). **OPEN — the Θ_ϕ-restricted case:** the witness is σ-DEcreasing while Θ_ϕ-reachable schedules are isotone (`multiFee_monotone`); the refutation settles only the general claim.

Definition 28 (Forward exchange function) [M7]. $\mathcal{S}(\Delta Q_M)$ is the output delivered against the money-leg input ΔQ_M at constant trading function — stated in Definition 12's signature (tick slot first, flow in the middle slot, L in the last slot; function signatures are never changed — user ruling 2026-08-04); the input rides the money leg of the flow, the output $\mathcal{S}(\Delta Q_M)$ is withdrawn from the asset leg (orientation per Definition 12, subject to the PR-ORIENT FLAG):

$$\varphi_{(\chi_{X/M},\, \epsilon_{X/M})}\Big(i(t);\; \big(\Delta Q_M,\, -\mathcal{S}(\Delta Q_M)\big),\; L\Big) \;=\; \varphi_{(\chi_{X/M},\, \epsilon_{X/M})}\big(i(t);\, 0,\, L\big)$$

$\mathcal{S}^{-1}$ is the reverse exchange function (CFMM_GEOMETRY); the source's glyph G enters as $\mathcal{S}$ (user ruling 2026-08-04 — sandwich semantics; also avoids any confusion with the G0–G6 block labels).

Definition 27 (Sandwich hazard) [M7]. Per MEV_THEORY_I eqs. (4)–(6) — the paper's slippage limit η enters as $\mathrm{tol}_{\mathrm{slip}}$ (η is the grid exponent, Definition 8; tol is the tolerance family), its user trade Δ is the money-leg flow ΔQ_M, and its `PNL` is the sandwich payoff π^{sandwich} (π-convention) . For a user trade ΔQ_M with slippage floor $(1 - \mathrm{tol}_{\mathrm{slip}})\mathcal{S}(\Delta Q_M)$, the front-run $\Delta Q_M^{\mathrm{sand}}(\Delta Q_M, \mathrm{tol}_{\mathrm{slip}})$ solves the slippage-binding equation, the back-run recovers the position, and the attacker's payoff is:

$$\begin{aligned}
\mathcal{S}(\Delta Q_M^{\mathrm{sand}} + \Delta Q_M) - \mathcal{S}(\Delta Q_M^{\mathrm{sand}}) \;&=\; (1 - \mathrm{tol}_{\mathrm{slip}})\, \mathcal{S}(\Delta Q_M) \\
\Delta Q_M^{\mathrm{sand}\prime} \;&=\; \Delta Q_M^{\mathrm{sand}} + \Delta Q_M - \mathcal{S}^{-1}\big(\mathcal{S}(\Delta Q_M + \Delta Q_M^{\mathrm{sand}}) - \mathcal{S}(\Delta Q_M^{\mathrm{sand}})\big) \\
\pi^{\mathrm{sandwich}}(\Delta Q_M, \mathrm{tol}_{\mathrm{slip}}) \;&=\; \Delta Q_M^{\mathrm{sand}\prime} - \Delta Q_M^{\mathrm{sand}} \;=\; \Delta Q_M - \mathcal{S}^{-1}\big(\mathcal{S}(\Delta Q_M + \Delta Q_M^{\mathrm{sand}}) - \mathcal{S}(\Delta Q_M^{\mathrm{sand}})\big), \\
\pi^{\mathrm{sandwich}}(\Delta Q_M, 0) &= 0
\end{aligned}$$

The **sandwich hazard** mirrors Definition 22's shape — extracted sandwich value per unit capital:

$$\lambda_{\mathrm{sandwich}}(t) \;\equiv\; \sum_{s<t} \frac{\pi^{\mathrm{sandwich}}\big(\Delta Q_M(s), \mathrm{tol}_{\mathrm{slip}}\big)}{\pi^{\mathrm{linear}}(s)} \;\geq\; 0$$

Under uniform batch clearing there is no ordering to exploit: $\Delta Q_M^{\mathrm{sand}} = 0 \;\Rightarrow\; \pi^{\mathrm{sandwich}} = 0 \;\Rightarrow\; \lambda_{\mathrm{sandwich}} = 0$ (the Angstrom regime). *Status:* **UNFORMALIZED** — no Lean carrier; the paper's profit bound (linear in $\mathrm{tol}_{\mathrm{slip}}$, with a liquidity hurdle) is cited, not transcribed. $\mathrm{tol}_{\mathbf{slip}}$ **— the conjecture is RESOLVED, split (`SandwichTol.lean`, CPMM member, all axiom-clean):**

- Θ_ϕ **branch REFUTED** (`sandwich_fee_hurdle_false`, 30 bp witness): the exact profitability frontier is $0 < \pi_\phi^{\text{sandwich}} \iff \phi(1-\phi)\,\Delta Q_M^{\text{sand}} < (1-\phi)(Q_M^L + \Delta Q_M) - Q_M^L$ (`pnlFee_pos_iff`) — $\mathrm{tol}_{\text{slip}}$ does not enter. The fee's true relationship pins an admissible TRADE SIZE, not the tolerance: $\Delta Q_M \le \frac{\phi}{1-\phi}\,Q_M^L \Rightarrow \pi_\phi^{\text{sandwich}} \le 0$ for every front-run (`sandwich_fee_hurdle_corrected`); above it, NO positive $\mathrm{tol}_{\text{slip}}$ closes the channel.
- Θ_p **branch PROVED as stated** (`sandwich_grid_cap`): within one spacing of the MARGINAL price (`priceRatio` $\le r$, $r = \lambda^{\eta\Delta_i}$ — the marginal-price step, the SQUARE of the grid step per Proposition 10), the binding tolerance is capped: $\mathrm{tol}_{\text{slip}} \le 1 - r^{-1}$.

So $\mathrm{tol}_{\text{slip}}$ is functionally BOUNDED by Θ_p and unconstrained by Θ_ϕ; it remains a free tolerance of the tol family inside the Θ_p cap. Supporting: `pnl_pos` (feeless sandwiches always profit), `slip_strictMono` (binding bijection), closed forms `slip_eq`/`pnl_eq`/`priceRatio_eq`/`pnlFee_eq` (Q_X^L cancels in every payoff).

Definition 23 (Aggregate MEV hazard) [M7]. The aggregate extraction hazard is the hazard-side sum

$$\lambda_{\text{MEV}} \equiv \lambda_{\text{ARB}} \oplus \lambda_{\text{sandwich}}$$

with $\oplus$ per Definition 19 / Theorem 14 ($\otimes_\phi$ acts on $[0,1]$, NEVER on unbounded hazards). $\lambda_{\text{sandwich}} = 0 \Rightarrow \lambda_{\text{MEV}} = \lambda_{\text{ARB}}$ — uniform clearing delivers this by construction, so Definition 22 through Theorem 19 transfer to λ_{MEV} verbatim in the Angstrom regime; the sandwich channel is a distinct object (MEV_THEORY_I), unmodelled here. **Protocol choice (DECIDED 2026-08-04):** MEV is controlled by the TAX — Rule 12's monoid entry (and the JIT liquidity tax when that section converts); auction-recycling mechanisms (ToB rebates) are NOT part of this protocol and are removed from this document — their formalized-not-adopted carriers remain in-tree. One parametric lever OUTSIDE Θ_ϕ:

Cadence = Δt: moves λ_{ARB} monotonically, absent from λ_{FLAIR} — the non-degenerate lever outside Θ_ϕ.

Caveats [M8] (annotations, no statement class): LEADING ORDER — everything rests on eq. (12) fast-block small-fee asymptotics; only Definition 22(ii), under its guard, is exact. QUASI-STATIC — $\mathbb{P}_{\Delta_{\text{ARB}}}$ is steady-state, applied per step on a σ-varying path (this document's extension; valid iff parameters are slow vs mispricing mixing). NO DEMAND ELASTICITY — the missing term is MMR §7.3 eq. (27); corner solutions are objective properties, NOT equilibrium claims (the elasticity layer is OPT_FEES). AGGREGATE SCOPE — two channels only; unmodelled: noise backruns, multi-block censoring (lengthens Δt), JIT (taxed separately), gas (additive fee). CADENCE VALIDITY — the Δt law is validated for block times $\gtrsim$ 1s; sub-second needs jump-diffusion, out of scope. The Theorem 19 σ-varying/Θ_ϕ-restricted split stands as labelled there.

Rule 12 (τ_{MEV} entry — monoid channel; DECIDED 2026-07-31) [M9]. The MEV tax enters the trader-paid fee through the proven Abelian monoid (Definition 17 / Theorem 14):

$$\phi_{\text{total}} \leftarrow \phi_M \otimes_\phi \phi_X \otimes_\phi \tau_{\text{MEV}}, \qquad \phi \otimes_\phi \tau_{\text{MEV}} \ge \phi \;\; (\tau_{\text{MEV}} \ge 0,\, \phi \le 1)$$

Alternates formalized, NOT adopted: (B) convex separation $\phi = (1-\tau_{\text{MEV}})\phi + \tau_{\text{MEV}}\phi$ (incidence-targeting, intensity-neutral); (C) auction lump-sum ToB recycling (`taxFraction`, `mevNet` — mechanism not part of this protocol; removed from the document 2026-08-04).

Theorem 20 (The discriminating algebra — what the monoid entry buys and cannot buy) [M10].

$$\begin{aligned}
\text{(A) intensity:}\quad & \mathbb{P}_{\Delta_{\text{ARB}}}\big(\phi \otimes_\phi \tau_{\text{MEV}}\big) \le \mathbb{P}_{\Delta_{\text{ARB}}}(\phi) \quad (\text{strict for } \tau_{\text{MEV}} > 0,\ \phi < 1)\\
\text{(A) no targeting:}\quad & (\phi_M \otimes_\phi \tau_{\text{MEV}}) \otimes_\phi \phi_X = \phi_M \otimes_\phi (\phi_X \otimes_\phi \tau_{\text{MEV}}) \quad (\text{aggregate leg-invariant})\\
\text{(A) hazard-exact:}\quad & (1-e^{-\lambda_M}) \otimes_\phi (1-e^{-\lambda_X}) \otimes_\phi (1-e^{-\lambda_\tau}) = 1 - e^{-(\lambda_M+\lambda_X+\lambda_\tau)}\\
(\text{A} \ne \text{B}):\quad & \exists\, \phi, \tau :\; (1-\tau)\big(\phi_M \otimes_\phi \phi_X\big) \ne \big((1-\tau)\phi_M\big) \otimes_\phi \big((1-\tau)\phi_X\big)\\
\text{(B breaks hazard):}\quad & \exists\, \tau, \lambda :\; 1 - e^{-\tau\lambda} \ne \tau\,(1-e^{-\lambda})
\end{aligned}$$

Consequences (proved): λ_τ is a genuine $\oplus$-summand (hazard-exact); the intensity effect is STRICT $\Rightarrow$ $\lambda_{\text{ARB}} \downarrow$; NO leg-targeting (benign flow pays); NO compensation routed (donation $\Rightarrow$ compose with (B)/(C), ORDER-SENSITIVE: tax-then-compose $\neq$ compose-then-split); $\phi \otimes_\phi \tau$ moves the level direction jointly ($\lambda_{\text{FLAIR}} \uparrow$, $\lambda_{\text{ARB}} \downarrow$).

Goal:

$$\boxed{\pi^\sigma \;=\; \Delta Q_v \left(\sigma^2(i(t)) - \sigma_K^2\right)^+}$$